Operating Systems - Orange Problem

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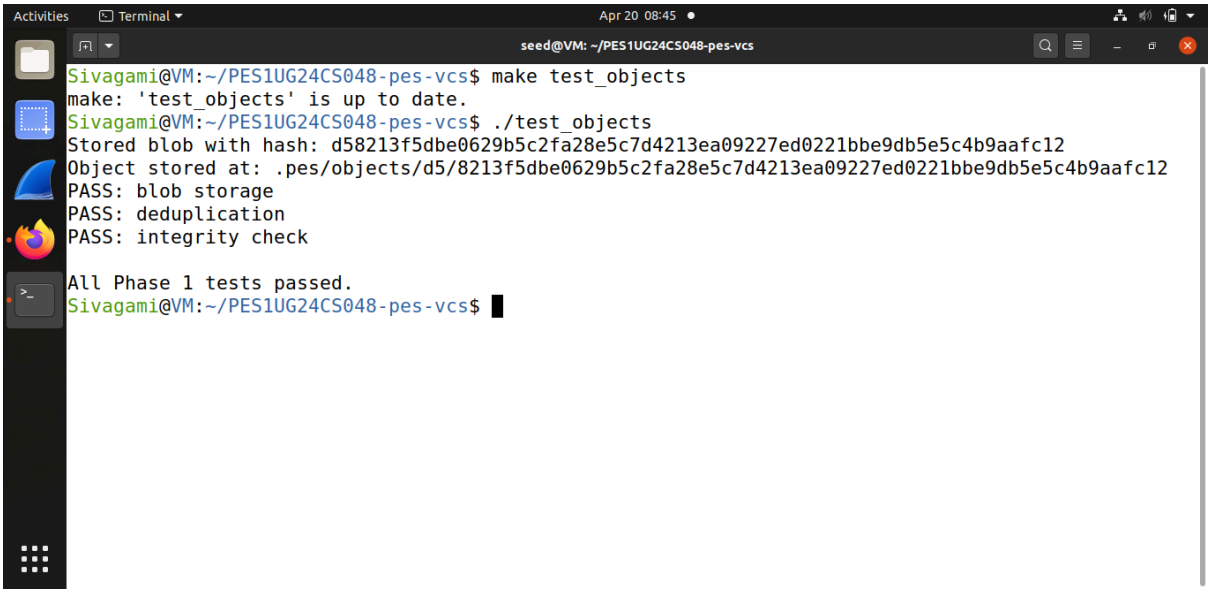
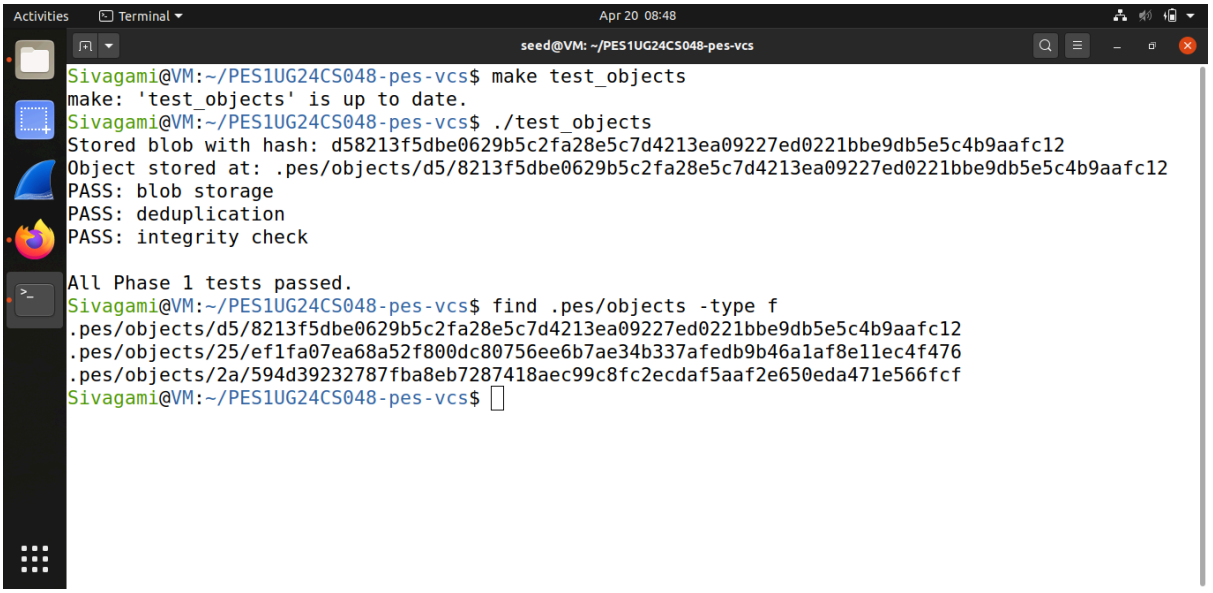
SEC: 4A

Phase 1: Object Strong Foundation

1A)



1B)



Phase 2: Tree Objects

2A)

```
Activities   Terminal   Apr 20 09:44
seed@VM: ~/PES1UG24CS048-pes-vcs
gcc -Wall -Wextra -O2 -c index.c -o index.o
index.c: In function 'index_save':
index.c:67:33: warning: format '%lu' expects argument of type 'long unsigned int', but argument 6
has type 'uint32_t' {aka 'unsigned int'} [-Wformat=]
   67 |         fprintf(f, "%o %s %lu %lu %s\n",
      |                                 ~~^
      |                                   |
      |                                   long unsigned int
      |                                 %u
......
   71 |             sorted.entries[i].size,
      |             ~~~~~~~~~~~~~~~~~~~~~~
      |                                 |
      |                                 uint32_t {aka unsigned int}
gcc -o test_tree test_tree.o object.o tree.o index.o -lcrypto
Sivagami@VM:~/PES1UG24CS048-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
Sivagami@VM:~/PES1UG24CS048-pes-vcs$
```

2B)

```
Activities   Terminal   Apr 20 09:48
seed@VM: ~/PES1UG24CS048-pes-vcs
      |                                 uint32_t {aka unsigned int}
gcc -o test_tree test_tree.o object.o tree.o index.o -lcrypto
Sivagami@VM:~/PES1UG24CS048-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
Sivagami@VM:~/PES1UG24CS048-pes-vcs$ find .pes/objects -type f
Sivagami@VM:~/PES1UG24CS048-pes-vcs$ mkdir -p .pes/objects
Sivagami@VM:~/PES1UG24CS048-pes-vcs$ echo "100644 0000000000000000000000000000000000000000000000000
0000000000000000 0 11 file.txt" > .pes/index
Sivagami@VM:~/PES1UG24CS048-pes-vcs$ ./run_tree
Segmentation fault
Sivagami@VM:~/PES1UG24CS048-pes-vcs$ find .pes/objects -type f
Sivagami@VM:~/PES1UG24CS048-pes-vcs$ printf "100644 file.txt\0" > tree.bin
Sivagami@VM:~/PES1UG24CS048-pes-vcs$ dd if=/dev/zero bs=32 count=1 >> tree.bin 2>/dev/null
Sivagami@VM:~/PES1UG24CS048-pes-vcs$ xxd tree.bin | head -20
00000000: 3130 3036 3434 2066 696c 652e 7478 7400  100644 file.txt.
00000010: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000020: 0000 0000 0000 0000 0000 0000 0000 0000  ................
Sivagami@VM:~/PES1UG24CS048-pes-vcs$
```

Phase 3: The Index (Staging Area)

3A)

3B)

Phase 4: Commits and History

4A)

```
Activities  Terminal  Apr 20 13:02
Terminal
Committed: bcce8c54e412... second commit
Sivagami@VM:~/PES1UG24CS048-pes-vcs$ echo "file3" > c.txt
Sivagami@VM:~/PES1UG24CS048-pes-vcs$ ./pes add c.txt
Sivagami@VM:~/PES1UG24CS048-pes-vcs$ ./pes commit -m "third commit"
Committed: 5931032778e7... third commit
Sivagami@VM:~/PES1UG24CS048-pes-vcs$ ./pes log
commit 5931032778e7fe318ed68aabe0b7f64885756c9a45f249f183fb84acd532b85d
Author: Sivagami <PES1UG24CS048>
Date:   118784

    third commit

commit bcce8c54e41211d76abb9f392678209e97260e6f56aaffaa49f3a21b979d409a
Author: Sivagami <PES1UG24CS048>
Date:   118784

    second commit

commit 96642b2bcfd02e700f304b6f55183f3a22ee784507a0b5e8cd540f6bd9edb1f4
Author: Sivagami <PES1UG24CS048>
Date:   118784

    first committ
```

4B)

```
Activities  Terminal  Apr 20 13:03
Terminal
Author: Sivagami <PES1UG24CS048>
Date:   118784

    first commit

Sivagami@VM:~/PES1UG24CS048-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/26/97019278243c1823b5ab720986ac5f2cd2083c3c2f4d29964c32daa7ed1b40
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/4f/31c356705a225cfb4416b6b011806a52104e4ff1d0a6029f7d9fab0bceb81a
.pes/objects/59/31032778e7fe318ed68aabe0b7f64885756c9a45f249f183fb84acd532b85d
.pes/objects/7b/6de9267ded53f053c975c2c6e9410b1a6a9deea8c27cbc37391b693aa9af07
.pes/objects/7b/cd3e2dc89ab1203c1e5cd2debaf00f3bfe798fd09876c5736d31f0a9ceb43f
.pes/objects/96/642b2bcfd02e700f304b6f55183f3a22ee784507a0b5e8cd540f6bd9edb1f4
.pes/objects/a7/3612171254d790a541c111e4856b978cf10cd2dc2ee414cb6cebc933580652
.pes/objects/bc/ce8c54e41211d76abb9f392678209e97260e6f56aaffaa49f3a21b979d409a
.pes/objects/cc/057ad4c4fdde5d92b27c0fb0fdc1c0a8564af60dbf6540a6661a26f5392896
.pes/objects/e1/292cfb746bbf0d03ee325f81aeec6129b10044052ad86ef03a2ed629b36b24
.pes/objects/f7/fd19bbbd1664265aeab1be3601b3decf85fe5bf96e7a95422ce70790ffaeaa
.pes/refs/heads/main
Sivagami@VM:~/PES1UG24CS048-pes-vcs$
```

4C)

```
Activities  Terminal  Apr 20 13:04
Terminal
Sivagami@VM:~/PES1UG24CS048-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
Sivagami@VM:~/PES1UG24CS048-pes-vcs$ cat .pes/refs/heads/main
5931032778e7fe318ed68aabe0b7f64885756c9a45f249f183fb84acd532b85d
Sivagami@VM:~/PES1UG24CS048-pes-vcs$
```

Phase 5: Branching and Checkout

Q5.1

checkout updates HEAD to point to branch ref, then rebuilds working directory from that commit tree; complex because it must safely overwrite files without losing local changes.

Q5.2

Compare working directory files with index + split diff of target branch tree; if index hash differs from current HEAD version → working tree is "dirty" and checkout is blocked.

Q5.3

Detached HEAD commits are stored but not attached to any branch; they can be recovered by creating a new branch pointing to that commit hash.

Phase 6: Garbage Collection

Q6.1

Mark-and-sweep: start from all branch heads → mark reachable objects → delete unmarked ones; use hash set for visited nodes; ~1–2 passes over all reachable commits/trees/blobs.

Q6.2

GC might delete objects not yet written by a concurrent commit → race condition; Git avoids this using locking + reachability snapshot during GC.